

Merging SaaS Products In A User-Centered Way — A Case Study of Overleaf and ShareLaTeX

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Figure 1: Most of the combined Overleaf development team meeting in person shortly after the merger was announced. Overleaf and ShareLaTeX were both fully remote teams, and bringing the two teams together was a key milestone in the merger.

ABSTRACT

Mergers and acquisitions pose complex organizational and technical challenges. A user-centered approach can help ensure the success of a merger. By prioritizing the management of user expectations, by addressing change aversion and by ensuring users feel heard, a merger cannot only be made more successful, but also be perceived as a positive change by end users. This case study describes and discusses the user-centered approach taken to successfully merge two established online LaTeX editors: Overleaf and ShareLaTeX. A mixed methods approach was used to simultaneously ensure broad coverage of the user base (quantitative; relevant because these were established products) and also in-depth understanding of their issues, opinions and motivations (qualitative; relevant to making informed design decisions). This user-centered approach was successful in helping the company to set a path for the merger, to adapt in the face of new data, and to create the foundation for sustained growth.

CCS CONCEPTS

• **Human-centered computing** → Empirical studies in HCI; HCI design and evaluation methods; Synchronous editors.

KEYWORDS

case study, latex editors, user-centered design, mixed methods research, mergers

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1 INTRODUCTION

LaTeX [10] is a typesetting system widely used in academia, and online LaTeX editors support the increasingly collaborative nature of academic writing. Overleaf (OL) version 1 (v1) and ShareLaTeX (SL) were, in 2017, two established online platforms for LaTeX editing, with over one million users collectively [12, 23]. Competing for the same user base, both companies saw sustained growth and had developed two independent platforms which provided essentially the same core functionality: online LaTeX editing and compiling. However, each had its own identity and specific strengths: OL v1 had rich-text/WYSIWYG (What You See Is What You Get) editing capabilities more welcoming to non-LaTeX users, and stronger

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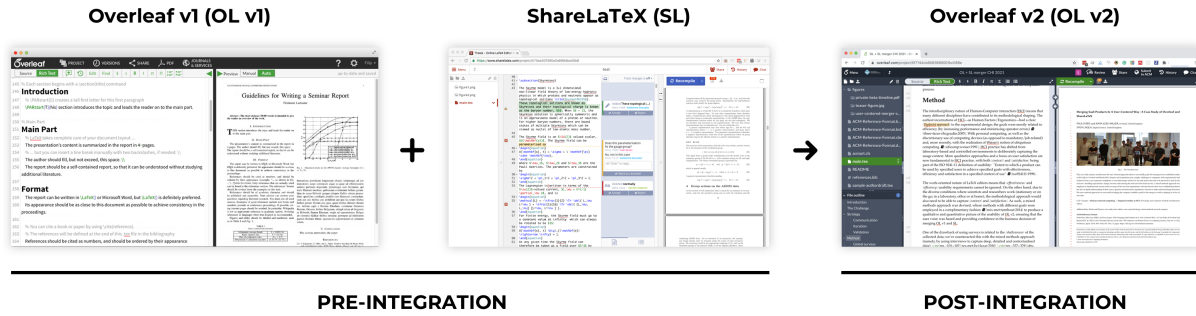


Figure 2: The legacy OL v1 and SL, followed by the new OL v2

partnerships with journals, publishers and publishing systems, including direct submissions from the editor; SL had a lower latency real-time editor with additional collaboration and reviewing features, including a tracked-changes mode.

In July 2017 [13], OL and SL announced that the companies were joining forces to work together on a combined platform. This happened through an acquisition; OL fully acquired the company behind SL. Having all members from both teams together and focusing on a single platform had the potential to avoid duplication of efforts in building, maintaining and improving the same core functionality, while sharing expertise on the specific strengths of each. A private beta (with three distinct phases) of the new combined platform started in January 2018; it was then followed by a public beta and, in September of the same year, the final launch of the new platform: Overleaf v2 (OL v2).

From a technical standpoint, the new OL v2 was built using components both from OL and SL. The LaTeX compiling infrastructure and real-time collaboration system were brought from SL; the version control, submission, and institutional affiliations system came from OL. Perhaps most relevant from a UX standpoint, the user-facing part (the web application) of the platform was based on the SL one. The application was redesigned with an updated OL brand and received significant work to implement features that existed in OL but not in the original SL web application (Fig. 2).

Software mergers are typically motivated by at least one of the following: to acquire new technology, to gain additional market share, to remove a competitor, or to bring expertise in-house. Mergers and acquisitions also come with a plethora of challenges - financial, technological, legal and human in nature [4, 9]. In particular, when two similar products are being merged — as described in this case study — a multitude of integration challenges present themselves:

- (1) How can technology be re-used and integrated?
- (2) How can staff be retained and kept motivated? [6]
- (3) How can end users be brought on-board with the changes?

This case study outlines and discusses our approach to bringing users of OL and SL on-board with the merger. It describes the user-centered methods employed to engage users while bringing together technology and staff of two distinct organizations and demonstrate how this merger not only maintained the existing combined user base but led to significant user growth of our online LaTeX editing platform.

2 STRATEGY

The challenge of merging two similar, yet distinct, established platforms was a large one, both from a software engineering and a user experience (UX) point-of-view. Both platforms had established user-bases, which have personally invested their time to become proficient with a particular online LaTeX editor. OL and SL users had already established habits and expectations from each of the platforms; some even tried both and made an explicit choice towards one or the other. An online LaTeX editor is a tool to achieve specific goals (writing a paper, dissertation, report) and any change to a tool is met with some justifiable aversion: if the user is productive and familiar with a given tool, we want them to perceive the benefits of any change as unequivocally superior to the short-term productivity losses associated with acknowledging and internalizing change.

The importance of the effective management of change aversion in the evolution of software is well-established [22]. Merging two existing public-facing platforms carried the significant risk of alienating users. Proactively managing users' change and loss aversion tendencies [25] was a priority. Users generally prefer no change or incremental change to large updates, and large updates must unequivocally justify the time spent by the user in re-learning and establishing new habits. The fact that this was essentially an acquisition also had potential to cause concern in some users due to diminished competition in the product space. In order to minimize uncertainty and make it clear that the new platform would be an improvement, a three-pronged UX strategy was put in place to essentially keep end-users in the loop and provide reassurance that their voices and concerns were heard (Fig. 3):

USER-CENTERED MERGER STRATEGY



Figure 3: User-centered Merger Strategy

Communication. A communication strategy with frequent blog posts (and also using other channels, e.g. social media and email) was put in place as a way of reducing uncertainty, doubt, and to manage users' loss aversion. Major blog posts provided a way for the user to be heard and react, typically via surveys.

Iteration. A strategy based on launching early and iterating quickly was put in place to help users cope with the changes. Initially, OL v1 features were brought into SL for validation by non-OL users. Afterwards, the initial versions of OL v2 were made available to end-users (initially via private, invite only beta; later with a public beta). Features were deployed to the OL v2 beta as they were developed for quick validation and feedback.

Validation. A mixed methods approach combining quantitative and qualitative research methods was put in place in order to collect feedback and validate the new platform. Specifically, online surveys, remote usability testing, and open-ended interviews were used to complement the release of each new iteration. This allowed continuous user feedback early on and throughout the development process.

3 METHOD

The interdisciplinary nature of Human-Computer interaction (HCI) means that many different disciplines have contributed to its methodological shaping. The earliest incarnations of HCI — as Human Factors/Ergonomics — had a clear *Taylorist approach* to the experimental design; the goals were mostly related to efficiency (by increasing performance and minimizing operator errors) [8]. With personal computing, as well as the discretionary use of computing devices (as opposed to mandatory/job-related) and, more recently, with the realization of Weiser’s notion of ubiquitous computing [27], HCI practice has shifted from laboratory-based and controlled environments to deliberately capturing the usage context. More qualitative approaches and a focus on user satisfaction are now fundamental to HCI practice, with both *context* and *satisfaction* being part of the ISO 9241-11 definition of usability: “Extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use” [5].

The work-oriented nature of LaTeX editors means that *effectiveness* and *efficiency* usability requirements cannot be ignored. On the other hand, due to the diverse conditions where scientists and researchers work (stationary or on the go; in a laboratory, office or at home), the methodological approach would also need to be able to capture *context* and *satisfaction*. As such, a mixed methods approach was devised (summarized in Fig. 4), where methods with different goals were employed in a complimentary fashion [26] to produce a qualitative and quantitative picture of the usability of OL v2, ensuring that the user voice was heard and providing confidence in the business decision of merging OL v1 and SL. To mitigate geographical biases (due to the global nature of our global user base), all methods assumed no face-to-face meetings — the *remote first* nature of the company also made this a natural choice.

One of the drawbacks of using surveys is related to the *shallowness* of the collected data; we’ve counteracted this with the mixed methods approach (namely, by using interviews to capture deep, detailed and contextualized data) [11, pp. 101–102] [7, pp. 327–329].

Particularly when compared with surveys, interviews are designed for deeper exploration of problems, by allowing the interviewer to direct the discussion, ask subsequent questions (*why?*) and gather context (*when?*, *how?*) [11, pp. 178–179] [7, pp. 129–131]. The complementary nature of the benefits of both methods (breadth

and depth in surveys and interviews, respectively) means that both can be used together in a mixed methods approach — identifying significant issues by broadly surveying the user base and making sure the understanding of said issues is correct and contextualized by interviewing selected users.

3.1 Global surveys

Surveys were a key part of our research strategy. OL v1 and SL were established platforms with large user bases, and surveys allowed us to get input from these users in a cost-effective way. A pre-integration survey and a post-integration survey were conducted. Both surveys shared a very similar structure to allow a quantifiable before and after comparison of user satisfaction. In addition, the pre-integration survey was designed to gauge user expectations going forward, while the post-integration survey captured users’ experiences after the integration.

Both surveys consisted of three parts. In the first part, users were asked how they felt about the integration on a 5-point Likert scale. In the second part, closed questions focused on profiling users according to their prior experience with either one or both OL v1 and SL as well their level of subscription (free, personal paid subscription, or part of a group or institutional subscription). This enabled the subsequent creation of cohorts for further analysis. In the third part, open-ended questions aimed to explore what users liked most in the existing two products and what their expectations were going forward. Due to the large number of responses, these were analyzed using text mining techniques.

3.2 Feature-specific interviews

LaTeX editing is the key feature of OL and, as a consequence, it is usually the focus of most of the feedback. There are however other features of the platform which are relevant to the overall UX. In particular, a better understanding of how users organize their projects in the dashboard, and how they version their projects in the project history was needed to inform relevant design decisions. Contextual semi-structured interviews allowed for the direction of the discussion (i.e. refocusing on the main subject), while still letting participants mention other topics.

3.2.1 Dashboard interviews. In both OL and SL a user dashboard provides users with an overview of their projects and gives access to project management functionality (create, delete, rename, etc). 11 interviews were conducted to validate the new post-integration dashboard with users from both cohorts (8 OL, 3 SL users) as the pre-integration dashboards had significant design and conceptual differences. A secondary goal was to better understand usage of the post-integration dashboard - focusing on users’ experiences, requests and personal organization strategies.

3.2.2 Project history interviews. As with the dashboard, project history functionality was already available in both products, but with different user interfaces (UI). 12 interviews (7 OL, 5 SL users) were conducted to validate the existing features in the legacy (OL v1 and SL) platforms.

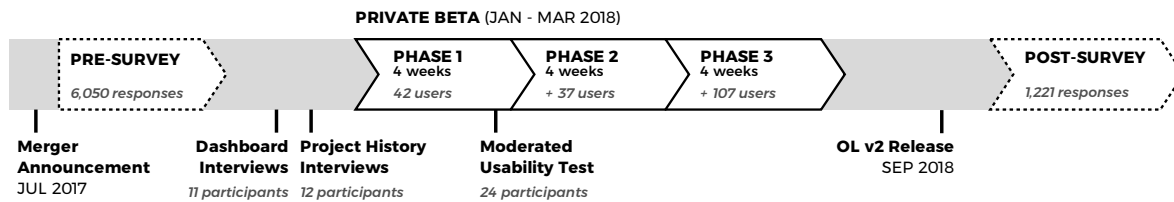


Figure 4: Merger timeline

3.3 Private beta

Frequent iteration was a fundamental aspect of our overall UX strategy, and the private beta was a key enabler of this. Allowing users to actually experiment with the product (as well as assessing and collecting feedback) was fundamental to ensure the ability to course-correct if needed. The private beta lasted roughly three months (Jan to Mar 2018), split into three phases of similar duration to allow for iterative feedback and opportunities for improvement to be captured. Each phase enlisted more users and exposed them to new features that were being rolled out as they were developed or migrated. As part of the private beta, a usability test was conducted.

3.3.1 Private beta — Phase 1. Users for phase 1 were onboarded via videoconferencing and screen-sharing by the UX team. As part of this onboarding meeting, a basic assessment of the usability of OL v2 was conducted to ensure a smooth start of phase 1. Phase 1 lasted 4 weeks and involved 42 users (29 OL, 13 SL). All participants were recruited from a pool of users who participated in earlier research initiatives. The over-representation of OL users in the sample was deliberate. As mentioned above, OL was closer to SL from a UX standpoint. There was a bigger risk of *friction* for users coming from OL, which was worth assessing.

3.3.2 Private beta — Usability test. At the end of phase 1, a validation [7, pp. 274] task-oriented and moderated usability test was conducted with 24 participants from the private beta phase 1 group (19 OL v1 users; 5 SL users; sampling bias was deliberate). Six tasks were developed to cover the basic functionality of OL v2 excluding LaTeX editing (to avoid conflation of OL v2 usability issues with general LaTeX usability issues). Each of the tasks had a list of items to look out for as well as exit questions. The concurrent think-aloud protocol was followed in order to capture the participants' feelings, thought processes and assumptions [1]; this made the conclusions more qualitative in nature while also providing context for some of the participants' struggles (while skewing quantitative aspects such as task completion times and success rates) [3].

3.3.3 Private beta — Phase 2. Phase 2 lasted 4 weeks and involved an additional 37 users (19 OL, 18 SL). These users were recruited from the respondents of the pre-integration survey. A *lighter* onboarding strategy was used, based on email — instructions and guidelines were conveyed via text (contrasting with the videoconferencing and screenshare sessions in Phase 1). Feedback was also requested and received via email.

3.3.4 Private beta — Phase 3. Phase 3 lasted 4 weeks and involved an additional 107 users (54 OL, 53 SL). Again, users were recruited

from the respondents of the pre-integration survey. As with Phase 2, email was used for onboarding and feedback collection.

4 RESULTS AND DISCUSSION

4.1 Global surveys

Overall, the results of the post-integration survey were positive when compared with the pre-integration survey. When users were asked about their feelings towards the integration, they rated it at 4.25 out of 5 post-integration, up from 4.11 pre-integration (an increase of 0.14, $p < 0.001$). Within this change in mean ratings, there are several noteworthy trends in the response rates and ratings for different groups.

The pre-integration and post-integration surveys received 6,050 and 1,221 responses, respectively. Several factors contributed to the larger number of pre-integration responses:

- The pre-integration survey was more widely circulated. It was part of the merger announcement, which was promoted via email to all users, prominent notices in both products, and social media (accounting for 35%, 55%, and 9% of responses, respectively, with the remaining 1% being of unknown origin). The post-integration survey was promoted only via in-product notices, so only users active during the survey saw it.
- The pre-integration survey collected responses for longer than the post-integration survey, at 43 days (20 Jul 2017 – 31 Aug 2017) and 17 days (8 Apr 2019 – 24 Apr 2019), respectively.
- The merger announcement generated more user interest and concern than the completion of the project.

Table 1 shows the number of responses to the pre- and post-integration surveys, classified according to the two profiling questions. The first profiling question asked which product(s) the user had used previously, *OL v1* only, *SL* only, or *both* products. In the pre-integration survey, users who had used SL only made up the largest group at 42% of respondents, compared to 24% who used OL v1 only and 34% who used both. In the post-integration survey, this reversed; users who had used OL v1 only made up the largest group at 32%, compared to 26% who used SL only and 29% who used both. This change likely reflects who was most concerned about the integration outcome when it was announced versus when it was completed — SL users before and OL v1 users after; more on this below. The remaining 13% of respondents in the post-integration survey answered that they had used neither pre-integration product (*None*), which is to say that they started using Overleaf after (or toward) the end of the integration.

Table 1: Number and percentage of responses to the pre- and post-integration surveys by the user’s subscription and the product(s) with which they had previous experience. There are fewer responses for the post-integration survey. SL users were the largest group in the pre-integration survey, whereas OL v1 users were the largest group in the post-integration survey.

		Previous Experience								
		OL v1		SL		Both		None	Total	
		Pre	Post	Pre	Post	Pre	Post	Post	Pre	Post
Subscription	Free	1157 (19%)	248 (20%)	1948 (32%)	217 (18%)	1599 (26%)	242 (20%)	124 (10%)	4704 (78%)	831 (68%)
	Personal	122 (2%)	62 (5%)	301 (5%)	42 (3%)	153 (3%)	56 (5%)	7 (1%)	576 (10%)	167 (14%)
	Group	163 (3%)	79 (6%)	316 (5%)	64 (5%)	291 (5%)	56 (5%)	24 (2%)	770 (13%)	223 (18%)
	Total	1442 (24%)	389 (32%)	2565 (42%)	323 (26%)	2043 (34%)	354 (29%)	155 (13%)	6050 (100%)	1221 (100%)

The second profiling question asked whether the user had a free account (*Free*), had purchased an individual subscription (*Personal*), or received a paid subscription as a member of a group or institutional license (*Group*). The proportion of responses across these groups remained relatively stable between the pre- and post-integration surveys. Overall, users with paid accounts were over-represented in the sample relative to the population, perhaps reflecting their greater engagement with the product and interest in the outcome of the integration.

How respondents felt about the integration on the Likert scale. A rating on the 5-point scale was provided in all 6050 pre-integration survey responses and 1050 (of 1221) post-integration survey responses. Respondents to the latter who had experience with neither pre-integration product or who did not answer the question are excluded from the analysis here. Table 2 shows the mean rating for each group and overall pre- and post-integration. The statistical significance of the change in mean rating for each group is assessed using a two-tailed, unequal variance (Welch) *t*-test and a significance threshold of $p < 0.05$.

As noted above, the overall mean rating increased by 0.14 from 4.11 to 4.25 ($p < 0.001$). Some increase might be expected due to survivor bias — users who were unhappy with the integration may have ‘voted with their feet’ and left, missing the post-integration survey. While it is difficult to control for this effect completely, it is notable that overall user numbers continued to increase [18–20], and that the rating increase was not uniform across all respondent groups.

One striking difference is in the *Previous Experience* groups. Ratings for SL users increased by 0.42 from 3.78 pre-integration to 4.20 post-integration ($p < 0.001$), whereas ratings for OL v1 users decreased by 0.23 from 4.40 to 4.17 ($p < 0.001$). The increase in SL users’ ratings suggests that they had concerns when the merger was announced but that these concerns were largely addressed. The decrease in OL v1 users’ ratings may reflect the fact that post-integration OL took many key parts of the application, mostly importantly the editor, from SL. This resulted in a smaller change for SL users and a larger change for OL v1 users, which was perhaps the opposite of what users expected given that OL acquired SL. Overall,

OL v1 user ratings decreased less than SL user ratings increased, and they converged toward a similar post-integration rating (4.17 and 4.20, respectively). Users who had used both pre-integration products saw no significant rating change.

The free text responses support this interpretation of the ratings. When, pre-integration, users were asked what we should ‘keep in mind for the integration’, the top 5 bigrams (after stopword removal and stemming with the Snowball list) for SL users were *open source*, *track changes*, *free account*, *wysiwyg editor*, and *user interface*. These speak to concerns about whether OL v2 would remain open source like SL (it did) and whether the new track changes feature and the free account would remain (they did). Sentiment about the OL WYSIWYG editor was mixed, with some hoping it would be in v2 and some hoping it would not (it was). And finally, some said they preferred the SL UI. When, post-integration, users were asked what they missed about the product they used pre-integration, OL v1 users’ responses were more diverse and less numerous than in the above, making the bigram analysis less useful, but some common themes emerged from reading the responses: concerns about aesthetic changes were common. There were also lingering concerns about a planned change to the OL v1 Git integration, which was reversed based on user feedback. We will return to this and the common user interface concerns below.

It is also notable that the rating changes were most pronounced among users with personal paid subscriptions. Paying SL users and OL v1 users had the largest change in rating, with an increase of 0.60 from 3.50 to 4.10 ($p = 0.001$) and a decrease of 0.51 from 4.48 to 3.97 ($p = 0.003$), respectively. This may be because these users had the most invested in the products, and so were more sensitive to any real or anticipated changes. Ratings changed less for respondents who received a paid account through a group or institutional subscription, who were perhaps less personally invested; overall ratings changed by 0.26 from 4.07 to 4.33 ($p < 0.001$) for these respondents, but this was primarily driven by the change for respondents with previous experience with SL only, which was strong across all subscription groups. There was no statistically significant change in ratings among free users, but it is worth noting that the

Table 2: Each cell shows the mean rating on the Likert scale (1 – 5; higher is better) for the corresponding group in the pre- and then post-integration surveys. The up or down arrow indicates an increase or decrease, respectively. Statistically significant ($p < 0.05$) changes are shown in bold with a double arrow. Ratings decreased for former OL v1 users but increased more for former SL users, leading to a net positive change overall. This change was driven in large part by increases in ratings among users with paid accounts.

		Previous Experience							
		OL v1		SL		Both		Total	
Subscription	Free	4.39 ↓	4.11	3.85 ↑	4.24	4.35 ↑	4.35	4.15 ↑	4.23
		-0.28		+0.39		+0.00		+0.08	
	Personal	4.48 ↓	3.97	3.50 ↑	4.10	4.03 ↑	4.58	3.84 ↑	4.22
		-0.51		+0.60		+0.55		+0.38	
Group		4.41 ↑	4.51	3.65 ↑	4.16	4.32 ↓	4.29	4.07 ↑	4.33
		+0.10		+0.51		-0.03		+0.26	
Total		4.40 ↓	4.17	3.78 ↑	4.20	4.32 ↑	4.38	4.11 ↑	4.25
		-0.23		+0.42		+0.06		+0.14	

increase of 0.08 from 4.15 to 4.23 was on the edge of significance with $p = 0.06$.

4.2 Private beta usability test

This test was designed to cover the basic functionality of OL v2 and, as such, didn't feature any complex LaTeX editing tasks. Rather, it was focused on project creation and management (sharing, organizing), basic editing, handling external files (figures) and compiling.

The most common issues for OL v1 users were related to changes in the UI (typically, items in the UI which have changed positions when compared to OL v1) and in default settings or *behaviors*. For example, the algorithm to determine when to automatically compile was subtly different, and automatic compiling was on by default in OL v1 but off by default in OL v2, which some users noted.

For SL users, the most common issues were related to graphical changes. The OL v2 editor, although very similar to SL from a layout standpoint, has a new color scheme which is noticeably darker — SL users asked for a lighter theme. Auto-compiling, which was new for SL users, was also mentioned and, like with OL v1 users, the timing to trigger a compile was explicitly mentioned, which indicated that the algorithm did not behave as users expected; it was later improved.

4.3 Interview Results

Both rounds of interviews surfaced some potential improvements to dashboards and the project history, but no major need for changes was identified. We found that interviewees were happy with whatever organizational strategies they employed to manage versioning, from using the existing project history interface built into OL to personal file/project-based strategies (e.g. creating duplicate versions of files or projects). More importantly, we found that users perceived both the dashboard and versioning to be minor parts of their overall experience (in contrast with the main document editing activities) and its implementation thus adequate for their needs. We were therefore confident in focusing our efforts in other areas.

4.4 Roadmap changes

Although the initial merger plan and scheduling were mostly followed, we did make changes to accommodate new requirements discovered through our ongoing communication and validation work. Two of the major changes were the reversal of a planned change to the OL v1 Git integration and a set of aesthetic changes to the editor *look & feel*: a light theme and configurable source-code font options.

The OL v1 Git integration. OL v1 and SL had two similar features designed to allow users to integrate their projects with Git, which is a widely used distributed version-control system. While SL provided an integration with a popular Git hosting provider, GitHub, OL v1 provided its own Git server, allowing users to interact with an Overleaf-hosted Git repository like any other git 'remote'. The SL feature was called GitHub sync, while the OL v1 one was called Git bridge.

The initial approach was to only add SL's GitHub sync to the new OL v2 — the burden of maintaining the two very similar features was assumed to be too high compared to the benefits, and GitHub sync was assumed to be more popular. However, data from the pre-integration global survey heavily suggested that the OL v1 Git bridge was fundamental to many users and their workflows, and that GitHub sync was not an adequate substitute. For example, a user in the pre-integration survey said we should keep in mind "The git support. In Overleaf v1 using it was very easy and it wasn't necessary to create a git hub [sic] repository." Some users raised privacy concerns: GitHub is a public-facing service, and repositories for free accounts were (at the time) public; it is also an external third-party having access to the Overleaf users' data (whereas Git bridge repositories were hosted by Overleaf). Some large institutional customers also noted that they had integrated the Git bridge into their internal systems.

In response to this user feedback, we decided to continue to support both GitHub sync and Git bridge, first through its own beta and finally rolled out to all users [14]. This episode illustrated how difficult it can be to remove features from a product, though we were successful in removing some other features that were either underutilized or made redundant by the integration.

A Light theme and configurable source-code font options. A significant portion of the feedback received from both OL v1 and SL users was related to color scheme and other user-interface choices. Although such issues might be classified as *cosmetic* in nature, they play a significant role in user satisfaction. Moreover, the multitude of different work contexts (physical characteristics of the environment as well as computing hardware, e.g. monitors) means that how the interface actually *looks and feels* to the user has some variability. This is particularly relevant in a LaTeX code editor, where typographical choices play a significant role in the editing experience (as well as in more physiological aspects such as eye strain). Both of these factors — user satisfaction and adaptability to different work environments — were deemed relevant and, as such, a set of interface customization options were added to accommodate personal preferences [17]. In particular, we added a light theme to the editor, as an alternative to the default OL v2 theme, which was darker than SL users were accustomed to, and we added additional options to the editor to allow the user to adjust the source code font and line spacing in the editor. These additional options in the product represent an ongoing burden that we will have to bear, but we decided that we were willing to pay that price in order to make more users feel ‘at home’ in the post-integration product. Nevertheless, some users still found they had to adjust; for example, in the post-integration survey, one SL user wrote “The change in ‘look’ has taken a while to get used to. Somehow things seemed less cluttered in the ShareLaTeX interface.”

5 CONCLUSION

The new OL v2 was launched in September 2018 with an email announcement and a blog post [16], followed by an official press release a few days later [15]. Although there’s no single number that validates the success of the merger, it has meanwhile reached significant usage milestones: from roughly one million users pre-integration in 2017, it reached 4 million users in March 2019 [19], 5 million users in November 2019 [18], and 6 million users in July 2020 [20].

The increasing usage numbers were also accompanied by some external recognition: In 2018, Overleaf was recognized as one of Britain’s Top 100 fastest growing businesses [21] and, in 2020, the company was a winner in the 2020 SaaS Awards Program in the category “Best SaaS for Nonprofits or Education” [2].

Internally, the merger was considered a success, with mixed methods-based user research being considered a fundamental driver behind it. As a result, user research has solidified its position as a defining aspect of the company *modus operandi*. Since the merger, Overleaf has employed various methods, ranging from ethnographic research to task-based usability tests. New features, validation of existing features and the main product roadmap are built on a solid understanding of user needs, both from quantitative and qualitative standpoints, as well as their usage contexts and typical tasks.

The communications component was effective overall in keeping users informed, but in hindsight there were opportunities to do more. One idea for practitioners facing similar challenges in the future would be to triangulate any impact of the communication strategy and participants’ engagement with it on the outcomes of

the various strands of research being conducted. Given the available resources at the time (and considerable uncertainty surrounding the General Data Protection Regulation, which took effect during the integration), we were not able to pursue this.

Assuming a more critical stance, it is at least arguable that — by merging Overleaf and ShareLaTeX — we have removed some competition, which might have contributed to the increase in usage numbers. However, further user research initiatives as well as constant monitoring of feedback (from various sources such as support inquiries and social media interactions) seem to paint a picture of an overall happy user-base, appreciative of the reliability and convenience of an online LaTeX editor with built-in support for collaboration. It is also worth mentioning that, since 2019, ACM CHI conference has been providing authors with a link to an Overleaf template for paper submissions [24]. This further contributes to the notion of Overleaf as a successful product — a sensible and popular choice for LaTeX-based authoring.

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